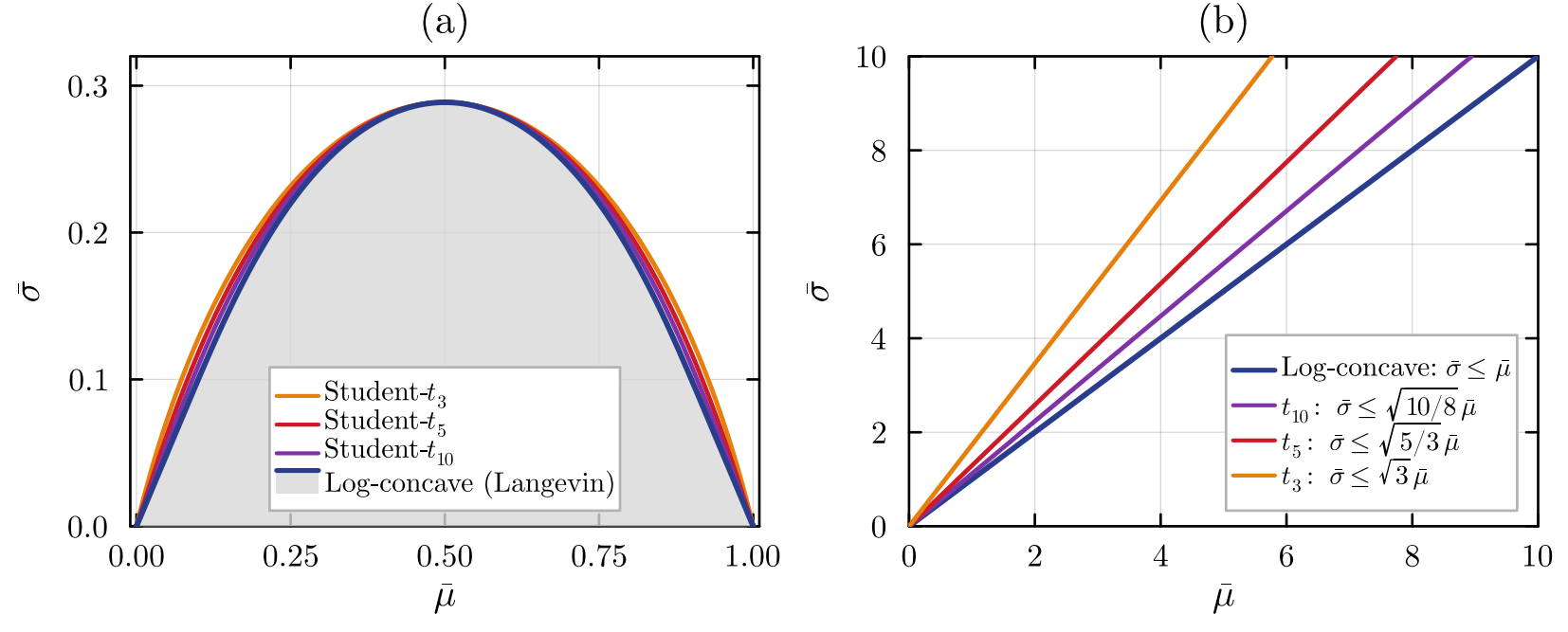
(a)
0.3
0.2
0.1
0.0
$\bar{\sigma}$
0.00
0.25
0.50
0.75
1.00
$\bar{\mu}$
Student-t_3
Student-t_5
Student-t_{10}
Log-concave (Langevin)
(b)
10
8
6
4
2
0
$\bar{\sigma}$
0
2
4
6
8
10
$\bar{\mu}$
Log-concave: $\bar{\sigma} \le \bar{\mu}$
t_{10}: $\bar{\sigma} \le \sqrt{10/8}\,\bar{\mu}$
t_5: $\bar{\sigma} \le \sqrt{5/3}\,\bar{\mu}$
t_3: $\bar{\sigma} \le \sqrt{3}\,\bar{\mu}$